

Algebraic Topology: a beginner's course

This lecture:

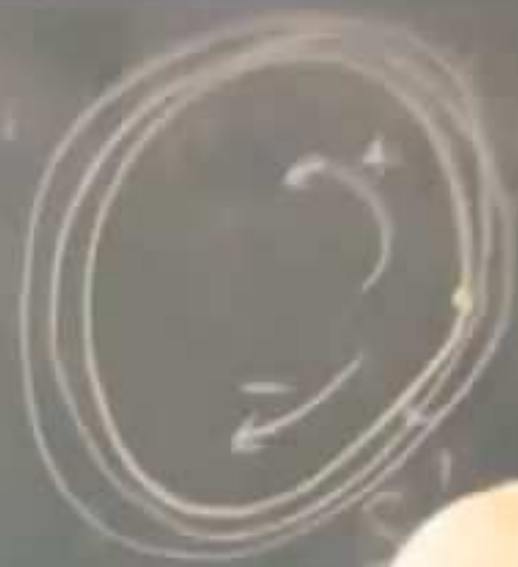
**AlgTop2: Homeomorphism and
the group structure on a circle**

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Lecture 2: Homeomorphism & the group structure on a circle.



More complicated paths:
Seems to be an outcome
for every integer



... $-2, -1, 0, 1, 2, 3, \dots$

Group structure on a level

F. Lehmermeyer ,

Gates

X_1, X_2, X_3 are collinear

parallel
lines
meet

B_1, B_2, B_3 are perspective from P .

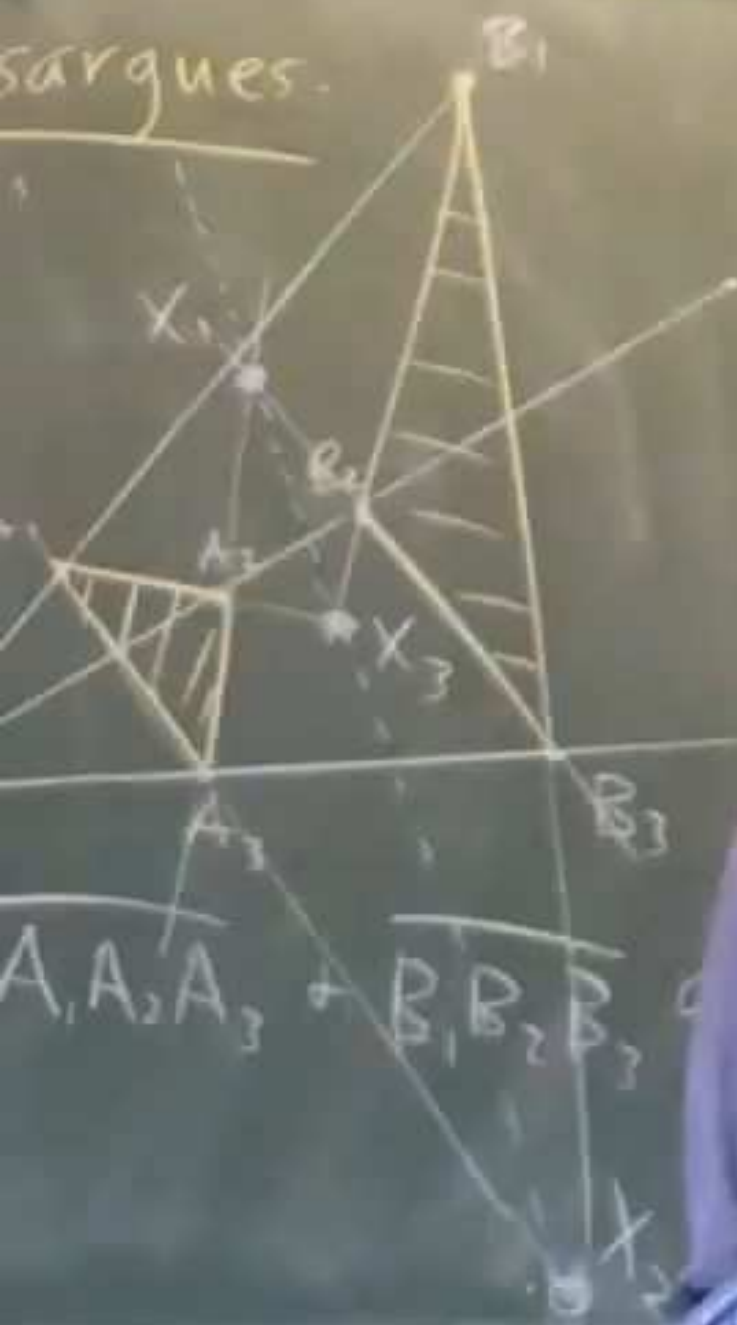
Pascal's thm. Pappus' thm extends
to case $A_1, A_2, A_3, B_1, B_2, B_3$ all
lie on a conic



order irrelevant
in context of
projective plane

$\approx$ Euclidean plane +
a "line at infinity"

sargues.

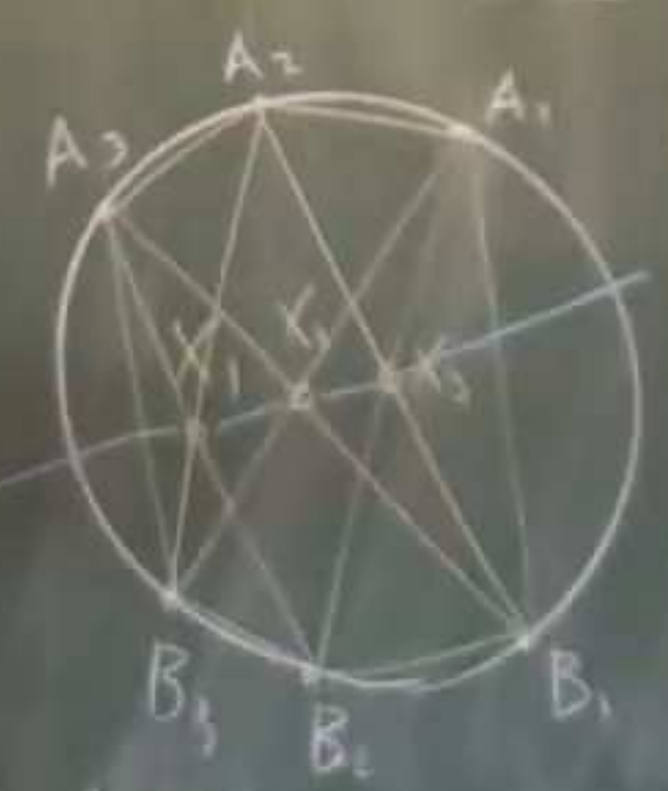


X_1, X_2, X_3 are collinear

parallel lines meet

$A_1, A_2, A_3 + B_1, B_2, B_3$

Pascal's thm Pappus thm extends to case $A_1, A_2, A_3, B_1, B_2, B_3$ all lie on a conic



order irrelevant in context of projective plane

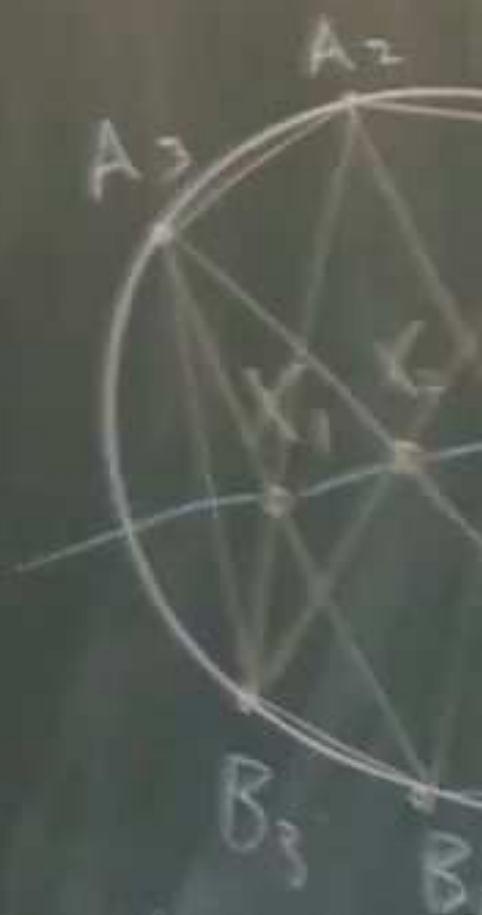
$\approx$ Euclidean plane + a line at infinity

line at
infinity

Group structure
exists on any conic

Prob 4 Show that with $e(h) = \left(\frac{1-h^2}{1+h} \right)$

Pascal's
to case
lie on



Prob 5 Show that with $e(h) = \left(\frac{1-h^2}{1+h^2}, \frac{2h}{1+h^2} \right)$

$$e(h_1) * e(h_2) = e(h_3) \text{ where}$$

$$h_3 = \frac{h_1 + h_2}{1 - h_1 h_2}$$

Prob 6 Exhibit the group structure on parabola $y = x^2$ with $0 = (0,0)$

$$\alpha(x) = [x, x^2]$$

Prob 6 Exhibit the group structure on hyperbola $y = \frac{1}{x}$ with

$$\beta(t) = \left[t, \frac{1}{t} \right]$$

Prob 4 Show that with $e(h) = \left(\frac{1-h^2}{1+h^2}, \frac{2h}{1+h^2} \right)$

$$e(h_1) * e(h_2) = e(h_3) \text{ where}$$

$$h_3 = \frac{h_1 + h_2}{1 - h_1 h_2}$$

Prob 5 Exhibit the group structure
on parabola $y = x^2$ with $0 = (0, 0)$

$$\alpha(x) \equiv [x, x^2]$$

Prob 6 Exhibit the group structure
on hyperbola $y = \frac{1}{x}$ with $0 = (1, 1)$

$$\beta(t) = \left[t, \frac{1}{t} \right]$$



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next in the series:

**AlgTop3:
Two-dimensional surfaces**